

Introduction to Probabilistic Graphical Models and Deep Generative Models

NOT-MIWAE: Deep Generative Modelling with Missing Not at Random Data

Matthieu Destrade
Félix Houdouin

December 17, 2024

1 Introduction

1.1 Overview

In this project we study the paper Ipsen, Mattei, and Frellsen 2021. The paper presents a new approach to deal with missing data in a dataset when missing patterns do not follow the Missing Completely At Random (MCAR) or Missing At Random (MAR) assumptions. We worked together on the understanding of the paper, Matthieu DESTRADE focused on the implementation of the model and section 3, and Félix HOUDOUIN on the first two parts of the report.

1.2 Presentation of the problem

Consider a data matrix $X = (x_1, \dots, x_n)^T \in \mathcal{X}^n$, whose rows are i.i.d. realizations of a random variable $x \in \mathcal{X} = \prod_{i=1}^p \mathcal{X}_i$, where $\mathcal{X}_i$ is the set of possible values for the i -th feature. We denote x_{ij} as the j -th feature of the i -th row of X . In our context, each row of the data matrix can be split into a missing part x_i^m and an observed part x_i^o .

The missingness pattern is described by a random variable $s \in \{0, 1\}^p$. We denote $S = (s_1, \dots, s_n)^T \in \{0, 1\}^{n \times p}$ as the mask matrix associated with X , defined by $S = (\mathbb{1}_{x_{ij} \text{ is observed}})_{1 \leq i \leq n, 1 \leq j \leq p}$. We decompose each realization of x into a missing part $x^m = x \odot (1 - s)$ and an observed part $x^o = x \odot s$, where $\odot$ denotes element-wise multiplication. We have the relationship $x = x^o + x^m$.

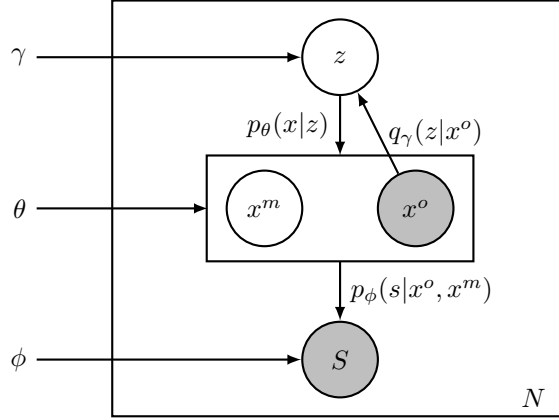
Our goal is to fill in the missing values in our dataset X , i.e., to construct a parametric model $p_{\theta, \phi}(x, s) = p_{\theta}(x)p_{\phi}(s | x)$ for the joint distribution. To tackle this task, several assumptions can be made:

1. **The Missing Completely At Random (MCAR)** assumption: Here, we assume $x \perp\!\!\!\perp s$, meaning that the missingness pattern does not depend on the values of the data. This implies $p(s | x) = p(s)$.
2. **The Missing At Random (MAR)** assumption: Here, we assume $s \perp\!\!\!\perp x^m$, meaning that the missingness pattern does not depend on the missing values. This implies $p(s | x) = p(s | x^o)$.

In this work, we do not make either of these assumptions and instead focus on the **Missing Not At Random (MNAR)** setting, for which no simplification can be made.

2 Presentation of the model

The paper presents an approach based on building and fitting deep latent variable models. The graphical model is the following:



The diagram of the model is presented in Figure 2.

2.1 The objective function

We want to maximize the log-likelihood of the observed data x^o given the parameters θ and ϕ of the model:

$$\ell(\theta, \phi) = \sum_{i=1}^n \log(p_{\theta, \phi}(x_i^o, s_i))$$

Intruducing the latent variable z_i we can rewrite the log-likelihood as:

$$\begin{aligned} p_{\theta, \phi}(x_i^o, s_i) &= \int p_{\theta, \phi}(x_i^m, x_i^o, s_i, z_i) dz_i dx_i^m \\ &= \int p_{\theta}(x_i^o | z_i) p_{\theta}(x_i^m | z_i) p_{\phi}(s_i | x_i^o, x_i^m) p(z_i) dz_i dx_i^m \\ &= \int \frac{p_{\phi}(s_i | x_i^o, x_i^m) p_{\theta}(x_i^o | z_i) p(z_i)}{q_{\gamma}(z_i | x_i^o)} q_{\gamma}(z_i | x_i^o) p_{\theta}(x_i^m | z_i) dz_i dx_i^m \\ &= \mathbb{E}_{z_i \sim q_{\gamma}(z_i | x_i^o), x_i^m \sim p_{\theta}(x_i^m | z_i)} \left[\frac{p_{\phi}(s_i | x_i^o, x_i^m) p_{\theta}(x_i^o | z_i) p(z_i)}{q_{\gamma}(z_i | x_i^o)} \right] \end{aligned}$$

Where we introduced the variational distribution $q_{\gamma}(z_i | x_i^o) = \Psi(z_i | g_{\gamma}(x_i^o))$ and the encoder g_{γ} , which aims at approximating the posterior $p(z_i | x_i^o)$ to reduce the variance of the Monte Carlo estimate of the intractable integral. Note that the neural network g_{γ} takes as input the observed part

of the data x_i^o , which has a constant shape based on our definition (i.e., $x_i^o = x_i \odot s_i$). Indeed, the missing part is set to zero. We suppose that $p(z_1) = \mathcal{N}(z_1; 0, 1)$.

We also assume that the model is fully factorized, meaning that the $x_{ij} \mid z_i$ are independent for all j . This assumption is not restrictive if we allow the latent variable z_i to have sufficient dimensions to adequately describe the data.

By denoting $w_{ki} = \frac{p_\phi(s_i | x_{ki}^o, x_i^m) p_\theta(x_i^o | z_{ki}) p(z_{ki})}{q_\gamma(z_{ki} | x_i^o)}$, we can rewrite the log-likelihood and introduce the not-missing importance-weighted autoencoder (not-MIWAE) lower bound $\mathcal{L}_K(\theta, \phi, \gamma)$:

$$\begin{aligned} \ell(\theta, \phi) &= \sum_{i=1}^n \log \left(\mathbb{E}_{z_i \sim q_\gamma(z_i | x_i^o), x_i^m \sim p_\theta(x_i^m | z_i)} \left[\frac{1}{K} \sum_{k=1}^K w_{ki} \right] \right) \\ &\geq \sum_{i=1}^n \mathbb{E}_{z_i \sim q_\gamma(z_i | x_i^o), x_i^m \sim p_\theta(x_i^m | z_i)} \left[\log \left(\frac{1}{K} \sum_{k=1}^K w_{ki} \right) \right] \\ &= \mathcal{L}_K(\theta, \phi, \gamma) \xrightarrow{K \rightarrow +\infty} \ell(\theta, \phi) \end{aligned}$$

This lower bound is the objective function we want to maximize during the training of the model.

2.2 Modeling $p_\phi(s \mid x)$ and $p_\theta(x \mid z)$

It remains to choose a parametric family for the conditional distributions $p_\phi(s \mid x)$ and $p_\theta(x \mid z)$. The distribution $p_\theta(x \mid z)$ should be fully factorized and belong to a reparametrizable family to allow computation of the gradient of the lower bound. If this is not the case, alternative methods can be used, but they often lead to increased complexity or higher variance in gradient estimation.

The missing data model $p_\phi(s \mid x)$ should be selected based on the missingness mechanism and be as specific as possible. In their paper, the authors chose a Bernoulli distribution for $p_\phi(s \mid x)$:

$$p_\phi(s \mid x) = \prod_{j=1}^p \pi_{\phi,j}(x)^{s_j} (1 - \pi_{\phi,j}(x))^{1-s_j}, \quad \text{where} \quad \pi_{\phi,j}(x) = \sigma(ax_j + b),$$

where a and b are learnable parameters of the model. A diagram of the resulting model, which is called nMIWAE-self_masking, can be found on figure 2.

The authors emphasize that this choice is not always optimal. For a missingness model invariant under some group action, a more specific approach should be employed by encoding this invariance into the model. Indeed, if the missingness model is too general, it becomes impossible to learn the true distribution of the data and the missingness mechanism. In fact, Molenberghs et al. 2008 shows that **MAR** (Missing At Random) and **MNAR** (Missing Not At Random) processes are empirically indistinguishable. Thus, using a **MNAR** model without prior knowledge of the missingness mechanism or on the data generation model leads to an inconsistent model.

3 Specification of the Model on a Homemade Dataset

3.1 The model

During the modeling of p_ϕ , the original model proposed in the paper makes two hypotheses:

$$\bullet \forall (i, j) \in \llbracket 1, d \rrbracket, i \neq j \implies (s_i | x) \perp\!\!\!\perp (s_j | x) \quad (1)$$

$$\bullet \forall (i, j) \in \llbracket 1, d \rrbracket, i \neq j \implies s_i \perp\!\!\!\perp x_j \quad (2)$$

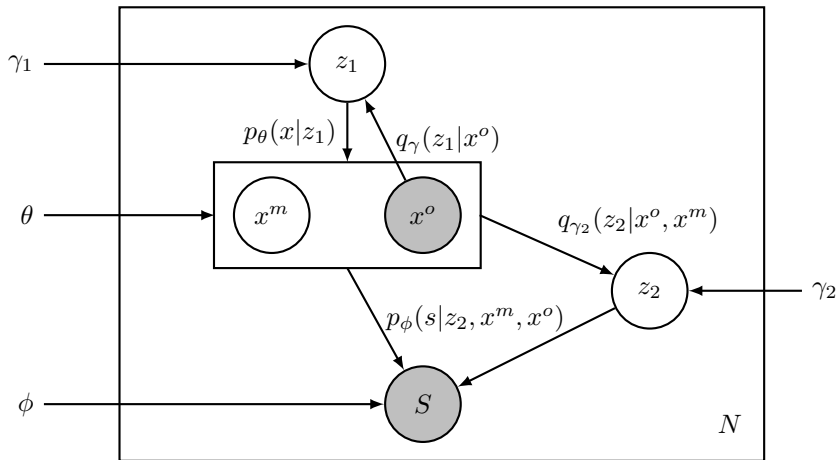
The paper proposes an agnostic approach in the experimental section that does not rely on the second hypothesis. This is achieved by computing the logits used to predict s with a dense layer rather than a linear regression on every component of x . To create a model that can handle data that do not respect the first assumption either, it is possible to introduce a latent variable z_2 into the model that directly influences s , as suggested in the conclusion of the paper. By doing so, we obtain:

$$p_\phi(s | x^o, x^m) = \int p_\phi(s, z_2 | x^o, x^m) dz_2 = \int p_\phi(s | z_2, x^o, x^m) p_2(z_2) dz_2$$

$$p_{\theta, \phi}(x_i^o, s_i) = \int \frac{p_\phi(s | z_2, x^o, x^m) p(z_1) p_2(z_2)}{q_\gamma(z_1 | x_i^o) q_{\gamma_2}(z_2 | x_i^o, x_i^m)} p_\theta(x_i^m | z_1) q_\gamma(z_1 | x_i^o) q_{\gamma_2}(z_2 | x_i^o, x_i^m) dz_1 dz_2 dx_i^m$$

which yields a new loss for our model.

We assume that $p(z_2) = \mathcal{N}(z_2; 0, 1)$ and $q_{\gamma_2}(z_2 | x) = \mathcal{N}(z_2; \mu_{\gamma_2}(x), \sigma_{\gamma_2}(x))$. This requires the use of a new encoder to learn $\mu_{\gamma_2}(x)$ and $\sigma_{\gamma_2}(x)$ from x^o and x^m . To model $p_\phi(s | z_2, x)$, we use the same approach as the one employed for the agnostic model in the paper. Specifically, we apply a dense layer to a concatenation of z_2 and x to compute the logits that define the Bernoulli distribution followed by s . The graphical model of such settings is:



A diagram of the model is also presented in Figure 3. Our model will be trained using an objective function derived in the same way as for the previously described nMIWAE model.

The dataset

To assess the capabilities of this new model compared to the original one, a specific dataset was designed. It is based on the white-wine dataset from the UCI database that was altered using an MNAR procedure. More precisely, samples for which the fixed acidity (x_1) is higher than the average (μ) were discarded with probability $q = 0.8$. For each of these samples, the values of volatile acidity (x_2) and citric acidity (x_3) were also discarded with a probability $0 < p < 1$. This allows obtaining a dataset that does not respect either of the hypotheses.

Indeed, we have:

- $\mathbb{P}(s_2 = 0, x_1 \geq \mu) = pq\mathbb{P}(x_1 \geq \mu)$ and $\mathbb{P}(s_2 = 0)\mathbb{P}(x_1 \geq \mu) = pq\mathbb{P}(x_1 \geq \mu)^2$
These two quantities are not equal since, in our case, $0 < \mathbb{P}(x_1 \geq \mu) < 1$. Therefore, x_1 and s_2 are not independent, and hypothesis (2) is not respected.
- $\mathbb{P}(s_2 = 0, s_1 = 1 \mid x_1 \geq \mu) = pq$ and $\mathbb{P}(s_2 = 0 \mid x_1 \geq \mu)\mathbb{P}(s_1 = 1 \mid x_1 \geq \mu) = pq^2$
These two quantities are not equal since $0 < q < 1$. Therefore, $s_2 \mid x$ and $s_1 \mid x$ are not independent, and hypothesis (1) is not respected.

Results

Using this dataset, we compared the agnostic model from the paper to our modified one (nMIWAE+latent). The metric that is used for comparisons is the Root Mean Squared Error (RMSE) of the reconstructed x^m data (scaled by the number of missing values in the dataset). It is imputed with:

$$\hat{x}^m = \sum_{k=1}^K \frac{w_k}{w_1 + \dots + w_K} \mathbb{E}(x^m | z_k)$$

for K samplings of z and x^m , where (w_k) is defined as previously. We also computed this metric for some baseline methods used in the paper (missForest, MICE, MIWAE and the self-masking nMIWAE model) for comparison.

Imputation RMSE	mean $\pm$ std
MIWAE	1.08 ± 0.02
nMIWAE self-masking	0.98 ± 0.03
nMIWAE agnostic	1.79 ± 0.03
nMIWAE + latent	1.04 ± 0.02
missForest	0.97 ± 0
MICE	0.95 ± 0

Table 1: Average results for 5 runs, $q = 0.8$ and $p = 0.8$

To better explore how the correlations between the missing data could affect the effectiveness of the methods, we tested them with different values of p .

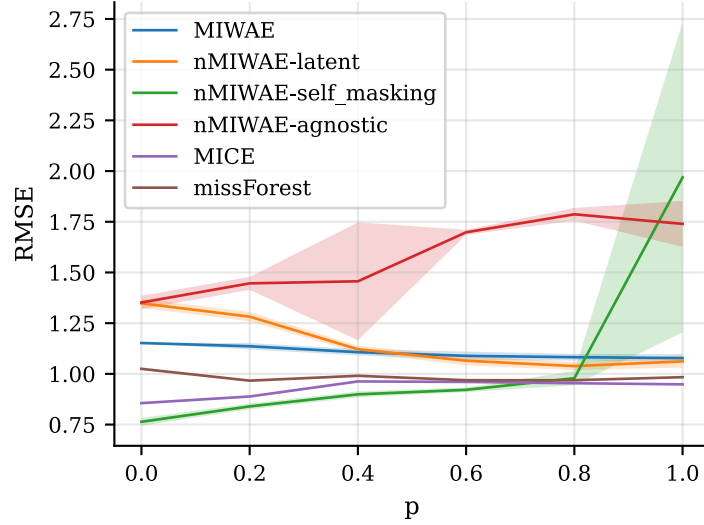


Figure 1: RMSE results depending on p for $q = 0.8$ along with their standard deviation over 5 generations

For $p = 0$, both hypotheses (1) and (2) are satisfied. We therefore expected the original network, nMIWAE-self_masking, to perform the best. For $p = 1$, none of the hypotheses are satisfied and, as anticipated, none of the original nMIWAE networks yields good results, and nMIWAE+latent outperforms nMIWAE-self_masking, nMIWAE-agnostic and the baseline MIWAE. However, all of these methods perform worse than the other baselines in these settings. We also explored the effectiveness of the models for $q = 1$, a slightly different setting in which only hypothesis (2) is not respected, and obtained similar results (see Figure 4).

4 Conclusion

To conclude, nMIWAE is a model that aims at learning the distribution of a dataset as well as the one of the missing data of this dataset, as this would allow for better results. While this works well in some particular cases, it appears that when the settings of the problem are more general, nMIWAE-based methods fail to perform better than simple methods such as MICE.

References

- Ipsen, Niels Bruun, Pierre-Alexandre Mattei, and Jes Frellsen (2021). *not-MIWAE: Deep Generative Modelling with Missing not at Random Data*. arXiv: 2006.12871 [stat.ML]. URL: <https://arxiv.org/abs/2006.12871>.
- Molenberghs, Geert et al. (Feb. 2008). “Every Missingness not at Random Model Has a Missingness at Random Counterpart with Equal Fit”. In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 70.2, pp. 371–388. ISSN: 1369-7412. DOI: 10.1111/j.1467-9868.2007.00640.x. eprint: https://academic.oup.com/jrsssb/article-pdf/70/2/371/49780193/jrsssb_70_2_371.pdf. URL: <https://doi.org/10.1111/j.1467-9868.2007.00640.x>.

Appendix

Diagram of the models

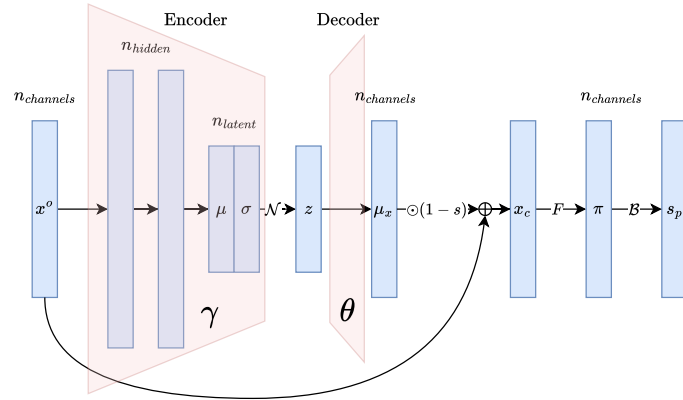


Figure 2: Diagram of the self-masking nMIWAE model
 (F is a dense layer for the agnostic model, and an affine transformation for the self_masking model)

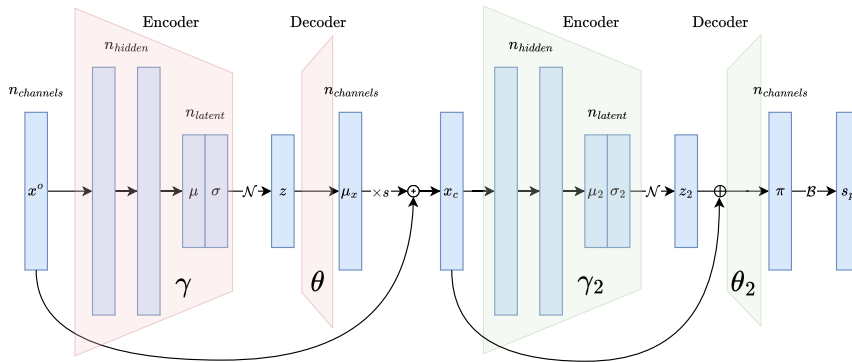


Figure 3: Diagram of the new nMIWAE + latent model

Implementation details

Our implementation of the models is based on the one from <https://github.com/nbip/notMIWAE> Ipsen, Mattei, and Frellsen 2021 and uses `tensorflow`.

The size of the latent spaces for z and z_2 are respectively $D - 1$ and D where $D = 11$ is the number of dimensions of the data used. The value for n_{hidden} is 128.

During training, for the loss computations, 20 Gaussian samples are sampled for z , and 1 for z_2 .

Additional results

Imputation RMSE	mean $\pm$ std
not MIWAE self-masking	1.90 ± 0.20
not MIWAE agnostic	1.29 ± 0.32
not MIWAE + latent	1.07 ± 0.11
missForest	1.18 ± 0
MICE	1.01 ± 0

Table 2: Average results for 10 runs, $q = 1$ and $p = 0.9$

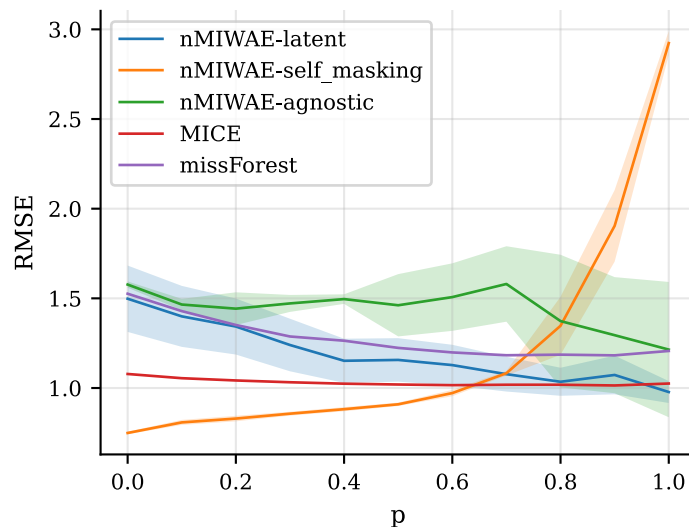


Figure 4: RMSE results for $q = 1$ along with their standard deviation over 10 generations

The nMIWAE-latent model used in this experiment is an alternative version of the one described in the report that supposes that $s|z_2$ is independent from x .